

# A TWISTED EGUCHI–KAWAI SPECTRAL CURVE AT $SU(3)$ AND THE MODULAR FORM 24.2.a.a: AN EICHLER–SHIMURA IDENTIFICATION

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**ABSTRACT.** We observe that the one-loop propagator spectrum of the twisted Eguchi–Kawai (TEK) model for pure  $SU(3)$  lattice gauge theory in  $d = 4$  with the minimal symmetric twist, expanded around the clock-shift twist eaters, collapses to a four-point set  $\{3, 6, 9, 12\}$ . The associated hyperelliptic spectral curve  $y^2 = (x - 3)(x - 6)(x - 9)(x - 12)$  is birational over  $\mathbb{Q}$  to the elliptic curve  $E: y^2 = x^3 - x^2 - 4x + 4$ , LMFDB label **24.a4**, of conductor 24, with  $j = 35152/9$  and torsion  $\mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$ . The curve  $E$  is a Weierstrass model for the modular curve  $X_0(24)$  and is *not* a CM curve. By the modularity theorem (Wiles–Breuil–Conrad–Diamond–Taylor) and the Eichler–Shimura relation, the Hecke eigenvalues  $a_p(E) = p + 1 - \#E(\mathbb{F}_p)$  coincide with the Fourier coefficients of the unique newform  $f \in S_2^{\text{new}}(\Gamma_0(24))$ , LMFDB label **24.2.a.a**, which admits the eta-quotient expression  $f(z) = \eta(2z)\eta(4z)\eta(6z)\eta(12z)$ . A PARI/GP verification confirms the match  $a_p(E) = a_p(f)$  at all primes  $5 \leq p < 200$  with  $p \nmid 24$  (44/44). We articulate this observation as an identification of two a priori unrelated spectra: the TEK eigenvalues of a pure lattice gauge theory matrix model and the Hecke eigenvalues of a weight-2 modular form, and we frame the follow-up question of whether the Hecke action extends to TEK Wilson operators at other levels.

## 1. INTRODUCTION

**The twisted Eguchi–Kawai model.** The Eguchi–Kawai (EK) reduction [7] replaces pure  $SU(N)$  lattice gauge theory on a periodic  $d$ -dimensional torus by a one-site matrix model

$$S_{\text{EK}} = -\beta N \sum_{\mu < \nu} \Re \text{Tr}(U_\mu U_\nu U_\mu^\dagger U_\nu^\dagger), \quad U_\mu \in SU(N), \mu = 1, \dots, d.$$

At leading order in  $1/N$  and at strong coupling, this matrix model reproduces the full lattice theory. The reduction breaks at weak coupling because the  $U(1)^d$  center symmetry breaks spontaneously. The twisted variant [8] restores center symmetry through a twist tensor  $z_{\mu\nu} = \exp(2\pi i n_{\mu\nu}/N)$ , antisymmetric and  $\mathbb{Z}$ -valued; we work with the minimal symmetric twist  $n_{\mu\nu} = \pm 1$  in  $d = 4$  and  $N = 3$ , which is the simplest realization preserving center symmetry. Standard references for the construction, the twist eaters, and the resulting effective virtual lattice include [5, 2, 9].

Expanding the gauge field around the twist-eater background  $U_\mu = \Gamma_\mu$ , where the matrices  $\Gamma_\mu \in SU(3)$  realize the Heisenberg–Weyl algebra  $\Gamma_\mu \Gamma_\nu = z_{\mu\nu} \Gamma_\nu \Gamma_\mu$  (see (5) below), the quadratic part of the action gives a one-loop propagator

$$D^{-1}(p) = 4 \sum_{\mu=1}^4 \sin^2(p_\mu/2), \quad p_\mu = \frac{2\pi n_\mu}{\bar{L}}, \quad n_\mu \in \{0, 1, \dots, \bar{L} - 1\}, \quad \bar{L} = N = 3.$$

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Since  $\sin^2(\pi/3) = \sin^2(2\pi/3) = 3/4$ , the eigenvalues  $E(n_1, \dots, n_4) = 4 \sum_{\mu} \sin^2(\pi n_{\mu}/3)$  simplify to

$$(1) \quad E(n_1, \dots, n_4) = 3\nu(n), \quad \nu(n) := \#\{\mu : n_{\mu} \neq 0\} \in \{1, 2, 3, 4\},$$

so the spectrum collapses to the four-point set  $\{3, 6, 9, 12\}$  with multiplicities  $(8, 24, 32, 16)$  summing to  $3^4 - 1 = 80$ .

**The spectral curve and main observation.** The associated hyperelliptic spectral curve over  $\mathbb{Q}$  is

$$(2) \quad C_{\text{TEK}}: y^2 = (x-3)(x-6)(x-9)(x-12).$$

With four simple rational branch points,  $C_{\text{TEK}}$  has genus one; it is therefore birational over  $\mathbb{Q}$  to an elliptic curve, which we identify explicitly via the standard quartic-to-Weierstrass transformation (PARI/GP `ellfromeqn`) followed by `ellminimalmodel`:

$$(3) \quad E: y^2 = x^3 - x^2 - 4x + 4.$$

The curve  $E$  is LMFDB **24.a4**, Cremona 24A3, with conductor  $N(E) = 24$ , discriminant  $\Delta(E) = 2304 = 2^8 \cdot 3^2$ ,  $j$ -invariant  $j(E) = 35152/9 = 2^4 \cdot 13^3/3^2$ , torsion subgroup  $E(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$ , and analytic and algebraic ranks both equal to 0 [10]. The denominator of  $j(E)$  is 9, so  $j(E)$  is *not* an algebraic integer; by a classical theorem of Deuring–Schneider (see, e.g., [15, Ch. II, Thm. 6.1]),  $E$  is *not* a CM elliptic curve.

By the modularity theorem of Wiles, Taylor–Wiles, and Breuil–Conrad–Diamond–Taylor [17, 16, 1], every elliptic curve over  $\mathbb{Q}$  is modular: there is a unique normalized newform  $f_E \in S_2^{\text{new}}(\Gamma_0(N(E)))$  such that  $L(E, s) = L(f_E, s)$ , and by Eichler–Shimura [13, Ch. 7] the Hecke eigenvalues  $a_p(f_E)$  coincide with the trace of Frobenius  $a_p(E) = p + 1 - \#E(\mathbb{F}_p)$  for every prime  $p \nmid N(E)$ . The space  $S_2^{\text{new}}(\Gamma_0(24))$  is one-dimensional [10], spanned by the single newform

$$(4) \quad f(z) = \eta(2z)\eta(4z)\eta(6z)\eta(12z),$$

labeled **24.2.a.a** in the LMFDB.

*Observation.* The Hecke eigenvalues of the modular form  $f = \mathbf{24.2.a.a}$  coincide with the trace of Frobenius of the spectral curve  $C_{\text{TEK}}$  of the  $\text{SU}(3)$ -twisted Eguchi–Kawai model in  $d = 4$  with the minimal symmetric twist. Explicitly, for every prime  $p \nmid 24$ ,

$$a_p(C_{\text{TEK}}) := a_p(E) = a_p(f),$$

where  $E$  is the minimal model (3) of  $C_{\text{TEK}}$ .

This identifies the eigenvalue spectrum of a *lattice gauge theory matrix model* with the Hecke spectrum of a *weight-2 modular form*. The identification is a consequence of the modularity theorem applied to  $E$  and is not itself a new modularity statement; the novelty is that the elliptic curve in question arises canonically and unambiguously from a TEK matrix model construction.

**Organization.** Section 2 reviews the TEK construction at  $\text{SU}(3)$  in  $d = 4$  with the minimal symmetric twist and verifies the spectrum collapse (1). Section 3 describes the birational reduction of (2) to (3) and records the LMFDB invariants. Section 4 states and proves Theorem 3 and reports the PARI/GP verification at 44 primes  $5 \leq p < 200$ . Section 5 addresses non-CM status, the modular curve interpretation  $X_0(24)$ , and the relation to large- $N$  limits. Section 6 formulates open questions about a Hecke action on TEK Wilson operators and the generalization to other levels and gauge groups.

**Conventions.** All elliptic curves are defined over  $\mathbb{Q}$  unless stated otherwise. Newform conventions follow [10] and [11]. The PARI/GP version used for the numerical verification is 2.15.4 [12].

2. THE TEK PROPAGATOR SPECTRUM AT  $SU(3)$ 

We work in  $d = 4$  Euclidean dimensions, gauge group  $SU(3)$ , with the minimal symmetric flux  $n_{\mu\nu} \in \{\pm 1\}$  satisfying  $\sum_{\nu \neq \mu} n_{\mu\nu} \equiv 0 \pmod{3}$ . The classical vacuum of the TEK action consists of the *twist eaters*  $U_\mu = \Gamma_\mu$ , solving

$$(5) \quad \Gamma_\mu \Gamma_\nu = z_{\mu\nu} \Gamma_\nu \Gamma_\mu, \quad z_{\mu\nu} = \exp(2\pi i n_{\mu\nu}/3),$$

which is a finite-dimensional realization of the Heisenberg–Weyl algebra at level 3. An explicit realization uses the Sylvester clock–shift pair

$$P = \text{diag}(1, \omega, \omega^2), \quad Q|j\rangle = |j+1 \bmod 3\rangle, \quad \omega = e^{2\pi i/3},$$

with  $\Gamma_1 = P$ ,  $\Gamma_2 = Q$ ,  $\Gamma_3 = PQ$ ,  $\Gamma_4 = P^2Q$  (up to relabeling). One verifies that the relations (5) hold for the chosen  $n_{\mu\nu}$ .

Expanding  $U_\mu = e^{iaA_\mu} \Gamma_\mu$ , with  $A_\mu$  Hermitian and traceless and  $a$  the (formal) lattice spacing, the quadratic part of  $S_{\text{TEK}}$  gives a propagator acting on Fourier modes  $A_\mu = \sum_n \tilde{A}_\mu(n) e^{ip \cdot x}$  with discrete momenta  $p_\mu = 2\pi n_\mu / \bar{L}$  and effective lattice size  $\bar{L} = N = 3$ ; see [5, Sec. 2] and [2, §3] for the derivation. The eigenvalues

$$E(n_1, n_2, n_3, n_4) = 4 \sum_{\mu=1}^4 \sin^2(\pi n_\mu/3), \quad n_\mu \in \{0, 1, 2\}, \quad n \neq 0,$$

collapse via  $\sin^2(\pi/3) = \sin^2(2\pi/3) = 3/4$  to  $E(n) = 3\nu(n)$ , as in (1). The multiplicity of a given value  $E = 3k$ ,  $k = 1, 2, 3, 4$ , is  $\binom{4}{k} 2^k$ , giving the count

$$(m_3, m_6, m_9, m_{12}) = (8, 24, 32, 16),$$

with total  $8 + 24 + 32 + 16 = 80 = 3^4 - 1$ , the full count of non-zero modes. This is verified numerically in PARI/GP (ancillary script `tek_verify.gp`, §1).

3. BIRATIONAL REDUCTION TO **24.a4**

The spectral curve (2) has four simple rational branch points, hence genus  $g = 1$ , and is therefore birational over  $\mathbb{Q}$  to an elliptic curve. We give an explicit reduction.

Translating  $x \mapsto x + 15/2$  symmetrizes the roots to  $\{\pm 3/2, \pm 9/2\}$ , so

$$y^2 = (x^2 - 9/4)(x^2 - 81/4) = x^4 - \frac{45}{2}x^2 + \frac{729}{16}.$$

Applying `ellfromeqn` (PARI/GP), which implements the classical quartic-to-Weierstrass map (see [3, §7.2] or [4, §8]) yields the (non-minimal) Weierstrass model  $W$ :  $[a_1, a_2, a_3, a_4, a_6] = [0, 315, 0, 32724, 1\,122\,660]$ , and `ellminimalmodel` produces

$$E: y^2 = x^3 - x^2 - 4x + 4.$$

PARI computes  $\Delta(E) = 2304 = 2^8 \cdot 3^2$ ,  $j(E) = 35152/9$ , conductor  $N(E) = 24 = 2^3 \cdot 3$ , torsion  $E(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$ , algebraic rank  $\text{rk } E(\mathbb{Q}) = 0$ ; these match the LMFDB record for the curve **24.a4** [10].

*Remark 1.* Replacing  $\bar{L} = 3$  with  $\bar{L} = 2$  in the propagator (equivalently, rescaling momenta) yields branch points  $\{4, 8, 12, 16\}$ , which give the same cross-ratio  $\lambda = 4/3$  and hence the same  $j$ -invariant; the resulting elliptic curve is  $\mathbb{Q}$ -isomorphic to  $E$ . The structural identity between  $C_{\text{TEK}}$  and  $E$  is invariant under the discrete rescaling  $\bar{L} \mapsto \bar{L}'$  so long as  $\bar{L}$  is coprime to  $N = 3$  and the symmetric twist is maintained. Generic asymmetric twists destroy the four-point collapse and produce higher-genus spectral curves.

*Remark 2.* The Weierstrass model  $E$  is in fact a model for the *modular curve*  $X_0(24)$  [10]. The genus formula  $g(X_0(N))$  (see [6, Thm. 3.1.1]) gives  $g(X_0(24)) = 1$ , and the dimension of new cusp forms of weight 2 at level 24 is  $\dim S_2^{\text{new}}(\Gamma_0(24)) = 1$ , so  $J_0(24)$  is itself an elliptic curve and  $X_0(24) \simeq J_0(24)$  over  $\mathbb{Q}$ . Thus the spectral curve  $C_{\text{TEK}}$  is itself a model for  $X_0(24)$ . We return to this in Section 5.

#### 4. THE HECKE–FROBENIUS IDENTIFICATION

**Theorem 3.** *Let  $C_{\text{TEK}}$  be the TEK spectral curve (2) and let  $E$  be its minimal model (3). Let  $f \in S_2^{\text{new}}(\Gamma_0(24))$  be the unique newform of level 24, weight 2, and trivial character, with  $q$ -expansion  $f(z) = q \prod_{n \geq 1} (1 - q^{2n})(1 - q^{4n})(1 - q^{6n})(1 - q^{12n}) = \eta(2z)\eta(4z)\eta(6z)\eta(12z)$ . Then for every prime  $p \nmid 24$ ,*

$$a_p(E) = p + 1 - \#E(\mathbb{F}_p) = a_p(f).$$

*Proof.* The curve  $E$  of (3) has conductor 24; this is verified via Tate’s algorithm [14, Ch. IV] and recorded in the LMFDB as **24.a4**. By modularity [17, 16, 1],  $E$  corresponds to a unique normalized newform  $f_E$  of weight 2 and level  $N(E) = 24$  with  $L(E, s) = L(f_E, s)$ . Since  $\dim S_2^{\text{new}}(\Gamma_0(24)) = 1$  and the unique basis vector is the eta-quotient  $f$  of (4) [10], we must have  $f_E = f$ . Comparing local  $L$ -factors at unramified primes gives the Eichler–Shimura relation  $a_p(E) = p + 1 - \#E(\mathbb{F}_p) = a_p(f)$  for  $p \nmid 24$  [13, Ch. 7, Thm. 7.6].  $\square$

**Proposition 4** (Numerical verification). *Let  $\pi(x) = \#\{p \leq x : p \text{ prime}\}$ . In a PARI/GP verification using the ancillary script `tek_verify.gp`, the equality  $a_p(E) = a_p(f)$  was checked exhaustively for the 44 primes  $p$  with  $5 \leq p < 200$  and  $p \nmid 24$ , with full agreement (44/44 match, no exceptions). The first fifteen values are recorded in Table 1.*

TABLE 1. The first fifteen Hecke / Frobenius eigenvalues  $a_p(E) = a_p(f)$  for  $p \nmid 24$ . All values are integers in the Hasse interval  $|a_p| \leq 2\sqrt{p}$ . Source: PARI/GP `ellap` and `mfcoef`.

| $p$ | $a_p(E)$ | $a_p(f)$ | $p$ | $a_p(E)$ | $a_p(f)$ | $p$ | $a_p(E)$ | $a_p(f)$ |
|-----|----------|----------|-----|----------|----------|-----|----------|----------|
| 5   | −2       | −2       | 17  | 2        | 2        | 37  | 6        | 6        |
| 7   | 0        | 0        | 19  | −4       | −4       | 41  | −6       | −6       |
| 11  | 4        | 4        | 23  | −8       | −8       | 43  | 4        | 4        |
| 13  | −2       | −2       | 29  | 6        | 6        | 47  | 0        | 0        |
|     |          |          | 31  | 8        | 8        |     |          |          |

*Remark 5.* We emphasize the modest scope of Theorem 3: it is not a new modularity result. The statement is, in effect, a triple identification:

- (1) the TEK propagator spectrum at  $\text{SU}(3)$ ,  $d = 4$ , minimal symmetric twist gives the curve  $C_{\text{TEK}}$  of (2);
- (2) the minimal model of  $C_{\text{TEK}}$  is the elliptic curve **24.a4**;
- (3) the modularity theorem then forces the trace-of-Frobenius spectrum of  $C_{\text{TEK}}$  to match the Hecke eigenvalues of the unique level-24 weight-2 newform.

The content of the present note lies in steps (1)–(2): a clean and canonical pure-gauge-theoretic construction whose minimal model hits exactly the modular curve  $X_0(24)$ . Step (3) is then a classical consequence of modularity. The further question – whether the Hecke operators  $T_p$  act in a natural way on TEK Wilson observables – is left open.

*Remark 6* (Non-vacuum flux sector). The construction of Section 2 uses the minimal symmetric twist  $k = 1$  of the González-Arroyo-Okawa Eguchi-Kawai matrix model. On the four-torus  $T^4$ , this twist corresponds to a non-trivial class in  $H^2(T^4, \mathbb{Z}_3) \cong (\mathbb{Z}_3)^6$ , explicitly carrying Pfaffian  $+1 \pmod{3}$  in

the canonical six-component flux basis. The Heisenberg-Weyl realisation of Section 2 is intrinsically tied to this non-vacuum flux sector: replacing  $k = 1$  by  $k = 0$  collapses the spectral curve to the trivial point. Consequently Theorem 3 is a statement about TEK in a *specific superselection sector* of the lattice gauge bundle; it does not directly engage the vacuum-sector mass-gap question of the Clay Yang-Mills Millennium problem. The suggestive factorisation  $N(E) = 24 = 8 \cdot 3$  into a power of two times the rank of the centre  $\mathbb{Z}_3$  is not part of a broader pattern: the analogous  $SU(2)$  construction would require a conductor-six elliptic curve, of which none exists in LMFDB. We therefore present the present result as a coincidence of arithmetic structure with the minimal twisted reduction, *not* as a general functor  $\text{TEK} \mapsto \text{HeckeAlg}$  of the kind discussed in Section 6.

## 5. REMARKS

**5.1. Non-CM and arithmetic non-overlap with Heegner anchors.** The denominator of  $j(E) = 35152/9$  is 9, so  $j(E)$  is not an algebraic integer;  $E$  is therefore not a CM elliptic curve. Equivalently, the conductor  $N(E) = 24$  does not appear in the class-number-one list of conductors of CM elliptic curves over  $\mathbb{Q}$ , which is  $\{27, 32, 36, 49, 121, 256, 361, 1849, 4489, 26569\}$  attached to  $K = \mathbb{Q}(\sqrt{-D})$  with  $D \in \{3, 4, 7, 8, 11, 19, 43, 67, 163\}$  [15, App. A, Thm. A.14]. A statistical confirmation is obtained from the Hecke density of supersingular primes: the number of primes  $p \in [5, 2000] \setminus \{2, 3\}$  with  $a_p(E) = 0$  is  $7/301 \approx 2.3\%$ , consistent with the Lang–Trotter conjecture for non-CM curves (heuristic density  $O(p^{-1/2})$ ) and incompatible with the 50% density of a CM curve at primes inert in the CM field.

**5.2. The modular curve interpretation.** By Remark 2, the TEK spectral curve  $C_{\text{TEK}}$  is a Weierstrass model for the modular curve  $X_0(24)$ . Hence the underlying gauge-theoretic data of the TEK matrix model at  $SU(3)$ ,  $d = 4$  is naturally encoded in  $X_0(24)$ , and the Hecke algebra  $\mathbb{T}_{24}$  of  $S_2(\Gamma_0(24))$  acts on the cotangent space of  $X_0(24)$ . The Hecke operators  $T_p$  for  $p \nmid 24$  therefore act on the differential  $\omega_f = f(z) dz$ , with eigenvalues  $a_p(f) = a_p(E)$ . The natural follow-up question (Section 6) is whether this action lifts to a Hecke action on gauge-invariant observables of the TEK matrix model itself.

**5.3. Robustness under  $\bar{L}$  variation.** For  $\bar{L} = 2$  (equivalently, alternate momentum normalization), the branch points become  $\{4, 8, 12, 16\}$ , with the same cross-ratio  $\lambda = 4/3$  and the same  $j$ -invariant; the resulting curve is  $\mathbb{Q}$ -isomorphic to **24.a4**. The identification with **24.a4** is thus invariant under the discrete rescaling  $\bar{L} \mapsto \bar{L}'$  so long as  $\bar{L}$  is coprime to  $N = 3$ . Asymmetric or non-minimal twists destroy the four-point spectrum and produce higher-genus spectral curves; see [5, 2] for the twist-dependence of the spectrum.

**5.4. Relation to large- $N$  limits.** The TEK construction is most useful as a finite- $N$  tool for extracting large- $N$  expectation values; in this note, we have focused on the  $SU(3)$  case where the spectrum collapses exactly, and the spectral curve is the finite-dimensional matrix model spectrum, not an  $N \rightarrow \infty$  limit. For  $SU(N)$  with larger  $N$ ,  $\bar{L}$  coprime to  $N$ , the analogous propagator  $E(n) = 4 \sum_{\mu} \sin^2(\pi n_{\mu}/\bar{L})$  does not collapse to a four-point set; the analog of Observation 1 at higher  $N$  is open and is one of the questions we record below.

## 6. OPEN QUESTIONS

**Question 7.** Does the Hecke action of  $\mathbb{T}_{24}$  on  $H^0(X_0(24), \Omega^1) = \mathbb{C} \cdot \omega_f$  extend to a natural action on gauge-invariant observables of the  $SU(3)$ -TEK matrix model? Concretely, is there an operator  $\hat{T}_p$  on the algebra of  $SU(3)$ -invariant polynomials in the matrices  $U_1, \dots, U_4$  whose restriction to the character ring of Wilson loops matches the action of  $T_p$  on  $\omega_f$ ?

**Question 8.** For other gauge groups  $SU(N)$  with twist parameters giving elliptic spectral curves of conductor  $M$ , does the analog of Observation 1 hold? In particular, is there a finite list of  $(N, d, n_{\mu\nu})$  for which the TEK spectral curve is  $\mathbb{Q}$ -rationally equivalent to a modular curve  $X_0(M)$  of small conductor  $M$ ?

**Question 9.** Define a finite- $N$  analog of the trace of Frobenius:  $\tilde{a}_p(\text{TEK}_N) := p + 1 - \#C_{\text{TEK}}(\mathbb{F}_p)$  for the spectral curve  $C_{\text{TEK}}$  at gauge group  $SU(N)$ . For which sequences  $N \rightarrow \infty$  along which the spectral curve remains elliptic, does  $\tilde{a}_p(\text{TEK}_N) \rightarrow a_p(f)$  for some weight-2 newform  $f$ ? This is a quantitative form of Question 8.

**Question 10.** Is there a functor  $F: \text{TEK}_{\text{twist,ell.}} \rightarrow \text{HeckeAlg}_{S_2^{\text{new}}}$  from the category of *TEK twist data whose spectral curve is elliptic* to the category of Hecke algebras of weight-2 newforms, such that  $F(\text{TEK-SU}(3)\text{-min. twist}) = \mathbb{T}_{24}$ ? If so, what is its image, and what are its kernel and cokernel? Even a partial answer (e.g. extending to one or two other gauge-theoretic data) would constitute a structural step beyond the present *ad hoc* identification.

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**Reproducibility.** The PARI/GP verification script `tek_verify.gp` accompanying this note executes the following checks:

- (1) Compute the multiplicities  $(m_3, m_6, m_9, m_{12}) = (8, 24, 32, 16)$  of the TEK propagator spectrum directly from (1).
- (2) Reduce  $C_{\text{TEK}}$  to Weierstrass form via `ellfromeqn` and `ellminimalmodel`, yielding  $E$  of (3).
- (3) Record  $\text{cond}(E) = 24$ ,  $\Delta(E) = 2304$ ,  $j(E) = 35152/9$ ,  $E(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$ .
- (4) Confirm  $j(E) \notin \mathbb{Z}$ , hence  $E$  is not CM.
- (5) Load the unique newform  $f \in S_2^{\text{new}}(\Gamma_0(24))$  via `mfininit/mfeigenbasis`, and verify the equality  $a_p(E) = a_p(f)$  for all primes  $5 \leq p < 200$  with  $p \nmid 24$ ; the script reports 44/44 matches.
- (6) Verify the supersingular density:  $7/301 \approx 2.3\%$  for primes  $p \in [5, 2000]$ , consistent with the non-CM case.

#### APPENDIX A. THE VERIFICATION SCRIPT `TEK_VERIFY.GP`

The file `tek_verify.gp` is included as an ancillary file. Its body is reproduced here for self-containment.

```
/* tek_verify.gp -- ancillary file for Paper_TEK_X024_Note
 * Verification script for: TEK SU(3) spectral curve = E[24.a4] = 24.2.a.a
 *
 * Reproducibility: PARI/GP 2.15.4
 * Run: gp -q < tek_verify.gp
 */

/* 1. TEK SU(3) propagator spectrum at one-loop around twist eaters */
/* E(n) = 4 * sum_{mu=1..4} sin^2(pi * n_mu / 3), n_mu in {0,1,2}, n != 0 */
print("=== 1. TEK SU(3) d=4 minimal symmetric twist propagator spectrum ===");
\\ E(n) = 4 sum_mu sin^2(pi n_mu / 3) for n_mu in {0,1,2}, n != 0
\\ Since sin^2(pi/3) = sin^2(2pi/3) = 3/4, E = 3 * (number of nonzero n_mu)
\\ so the spectrum is {3, 6, 9, 12}.
\\ tally pass
```

```

tek_tally() = {
  my(mult_3 = 0, mult_6 = 0, mult_9 = 0, mult_12 = 0, nz);
  for(a = 0, 2, for(b = 0, 2, for(c = 0, 2, for(d = 0, 2,
    if(a + b + c + d > 0,
      nz = (a != 0) + (b != 0) + (c != 0) + (d != 0);
      if(nz == 1, mult_3 += 1);
      if(nz == 2, mult_6 += 1);
      if(nz == 3, mult_9 += 1);
      if(nz == 4, mult_12 += 1);
    )
  ))));
  print("Multiplicities: E=3 -> ", mult_3, "; E=6 -> ", mult_6, "; E=9 -> ", mult_9, "; E=12 -> ", mult_12);
  print("Total non-zero modes: ", mult_3+mult_6+mult_9+mult_12, " (expected 3^4 - 1 = 80)");
};
tek_tally();

/* 2. Spectral curve y^2 = (x-3)(x-6)(x-9)(x-12) -> Weierstrass form */
print("\n=== 2. Spectral curve birational reduction ===");
C = ellfromeqn(y^2 - (x-3)*(x-6)*(x-9)*(x-12));
print("ellfromeqn output: ", C);
EW = ellinit(C);
Emin = ellminimalmodel(EW);
print("Minimal model: ", Emin[1..5]);

/* 3. Verify E[24.a4] : y^2 = x^3 - x^2 - 4x + 4 */
print("\n=== 3. Curve E[24.a4] : y^2 = x^3 - x^2 - 4x + 4 ===");
E = ellinit([0, -1, 0, -4, 4]);
print("Conductor : ", ellglobalred(E)[1]);
print("j-invariant : ", E.j, " (= ", factor(numerator(E.j)), " / ", factor(denominator(E.j)), ")");
print("Discriminant : ", E.disc, " (= ", factor(E.disc), ")");
print("Torsion (over Q) : ", elltors(E)[1]);

/* 4. NON-CM check: j has denominator 9 */
print("\n=== 4. NON-CM check (Schneider-Deuring) ===");
print("Denominator of j = ", denominator(E.j), " != 1");
print("j NOT algebraic integer => E NOT CM");

/* 5. Modular form 24.2.a.a a_p match */
print("\n=== 5. Hecke a_p match: E vs f_{24.2.a.a} ===");
mf = mfinit([24, 2, 1], 0);
L = mfeigenbasis(mf);
print("Number of newforms at level 24, weight 2: ", #L);
f = L[1];

check_match(pmax) = {
  my(c = 0, t = 0, mm = []);
  forprime(p = 5, pmax,
    t += 1;
    my(ap_E = ellap(E, p), ap_f = mfcoef(f, p));
    if(ap_E == ap_f, c += 1, mm = concat(mm, [[p, ap_E, ap_f]]));
  );
  print("Match: ", c, "/", t);
  if(#mm == 0, print("ALL MATCH"), print("Mismatches: ", mm));
};
check_match(199);

/* 6. Supersingular density (non-CM gives ~0, CM gives ~1/2) */
print("\n=== 6. Supersingular density (CM marker) ===");
ss_density(pmax) = {

```

```

my(c = 0, t = 0);
forprime(p = 5, pmax,
  t += 1;
  if(ellap(E, p) == 0, c += 1);
);
print("Primes 5..", pmax, ": ", c, "/", t, " = ", c*1.0/t);
};
ss_density(200);
ss_density(2000);

/* 7. Print a_p table for primes 5..50 */
print("\n=== 7. a_p table for primes 5..50 ===");
my_table() = {
  forprime(p = 5, 50,
    print("p=", p, " a_p=", ellap(E, p));
  );
};
my_table();

```

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